

Supplementary Methods and Reproduction Notes

Actual Participation Before Counterfactual Capacity

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S1. Scope

This supplement records finite formal witnesses and reproducibility notes for the main theoretical preprint. It contains no new human or animal data. The examples are analytic or finite computational constructions and do not establish phenomenal consciousness in any biological or artificial system.

S2. R125 finite trajectory witness

Six ternary registers evolve modulo 3:

$$\begin{aligned}a^+ &= a + b + gu, & b^+ &= a + 2b, \\c^+ &= c + d + gv, & d^+ &= c + 2d, \\u^+ &= u + v + a, & v^+ &= u + 2v + c.\end{aligned}$$

For fixed $g = 0$ and $g = 1$, exhaustive enumeration of all $3^6 = 729$ initial states gives:

T	0	1	2	3	4	5
identical initial-state trajectories	729	81	9	1	1	1

The result illustrates trajectory insufficiency.

S3. Probe witness

From the zero background with $a_0 = 1$,

$$x_1 = (1, 1, 0, 0, 1, 0).$$

Under $G_t \sim \text{Bernoulli}(\lambda)$,

$$D_3(\lambda) = \lambda.$$

This is a physical-response witness only; the paper does not infer a phenomenal law directly from this quantity.

S4. Symmetry witness

For the cycle graph C_5 , the automorphism group has order 10. The five adjacent two-node candidate subsets form one orbit, and no candidate is fixed by all automorphisms. This supports the Equivariant Selection Obstruction without implying that exchangeable candidates are numerically identical subjects.

S5. Claim-status discipline

PRD is a postulate. Idle-Support Invariance is conditional on PRD. Trajectory Insufficiency is an existence result. Bridge Underdetermination and Forced Bridge Factorization are formal limit results. The Equivariant Selection Obstruction is secondary. The silent-versus-idle unit contrast is an empirical research program.

S6. Publication boundary

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